Predictive Analytics and Recommender Systems

Recommender Systems - Association Rules and Content Based Filtering

Prakash B. Pimpale

June 15-17, 2017

1 Associations and CBR

1.1 Overview

- Introduction
- Association Rules
  - Apriori
  - FP-growth
- Content Based Filtering

1.2 Introduction

- An user reaches a product, service or information through : Search or Recommendation
- e-Commerce websites, course websites, Super markets, News paper portals
  - Recommend Explicitly - They directly say it - Ex. "Here are recommended products for you"
  - Recommend Implicitly - They indirectly prompt you - Ex. Place them near each other in store or website

- Various techniques and their variants : personalized or non-personalized
- Collaborative Filtering - Association Rule mining on historical data - Based on concept that certain things go together (Milk-Bread)
- Content Based Filtering - Based on content of items and user liking for those content
- Important Entities :
  - ITEMS : Things that are being bought - Movies, Books, Grocery items(milk, paste, etc.)
  - USER : The person whom things are to be recommended

1.3 Association Rules

- A typical purchase from a super market

(c) 2017 CDAC, Mumbai. All Rights Reserved

```
1. Rice
2. Milk
3. Bread
4. Eggs
5. Butter
6. Onions
7. Haldi
8. Mirch Powder
9.  ....
```

- For 1000 transactions at a supermarket - {Milk, Bread:400}, {Eggs, Onions, Haldi, Mirch Powder:250}, etc. are associated with each other
- A customers buying milk often buys Bread, or a customer buying TV often buys Antenna
- Famous example : 'Young' customers in 'urban region' buying Diapers were found often buying Beer
- Unexpected associations can come out - May help us increase sales
- Mining data from transaction tables, databases to get associations - Association Rule Mining
- Association Rules - {Eggs, Onions, Haldi} => {Mirch Powder}
- Associations can be used for - Discounting Strategy, Product placements, Finding loan defaulters, Diseases, etc.

1.3.1 Terms

- Transaction Database : Association Rules are learned from historical transactions/database

Transaction ID	Items
T1	Milk,Bread,Onion, Mirch Powder
T2	Bread, Onion, Mirch Powder
T3	Milk,Bread
T4	Milk,Onion, Mirch Powder
T5	Mirch Powder
T6	Milk,Bread,Onion, Mirch Powder

Transaction ID	Milk	Bread	Onion	Mirch Powder
T1	1	1	1	1
T2	0	1	1	1
T3	1	1	0	0
T4	1	0	1	1
T5	0	0	0	1
T6	1	1	0	1

Figure 1: Sample Data

- Association Rule : Representation

```
                LHS            =>     RHS
{Eggs, Onions, Haldi} => {Mirch Powder}

Antecedent or left-hand-side (LHS) : {Eggs, Onions, Haldi} - X
Consequent or right-hand-side (RHS) : {Mirch Powder} - Y
X => Y or X -> Y : Association Rule. Read as "X leads to to Y" or "those who X also Y"
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

- Support : Proportion in which the item or combination of items occure in transactions/databases

```
supp (Onion) = No. of Transactions with Onions in it/ Total no. of transaction
             = 3/6 = 0.5
or for a set of items, supp( {Onion,  Mirch Powder}) = 3/6 = 0.5
or for a rule, supp({Onion} => {Mirch Powder}) = 3/6 = 0.5
```

- Confidence : Proportion of the association to the LHS/X i.e proportion in which the rule holds true

```
conf({Onion} => {Mirch Powder}) = supp({Onion} => {Mirch Powder})/supp({Onion})
                                = 3/3 = 1
(Implies all the times onions were purchased Mirch powder was purchased)

Simplified :
conf(rule) = supp(rule)/supp(X)
```

- Lift : It's confidence of rule over support of RHS/Y. Support of Y is general purchase of the Y including the one with X

```
 lift({Onion} => {Mirch Powder})  = conf({Onion} => {Mirch Powder})/supp({Mirch Powder})
                                  = 1/ (5/6) = 1/0.833 = 1.200

(Implies that the Mirch Powder (with onion) will be purchased 1.2 times more likely
than the general purchase of it)

Simplified:
lift(rule) = conf(rule)/supp(Y)
```

- Various algorithms can be used to mine high frequent rules as per the threashold support and confidence

1.3.2 Apriori Algorithm

- A widely used algorithm for association rule mining
- Finds all itemsets that have minimum support (provided by us) - High frequent 'itemsets'
- And then Generates rules using those itemsets and minimum confidence threashold

```
Min Support = 0.5 (3 or more times in 6 transactions)
Min Confidence = 0.70
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

```
Frequent Itemset will be
{Milk}, {Bread}, {Onion}, {Mirch Powder}, {Milk, Bread}, {Milk, Mirch Powder},
{Bread, Mirch Powder}, {Onion, Mirch Powder}

Those who are excluded : {Bread,Onion}, {Milk, Onion}, {Milk, Bread, Onion, Mirch Powder}
for lack of sufficient support

Rules Generated :                      Support    Confidence Lift
{Bread}              => {Milk}         0.5000000 0.7500000  1.125
{Mirch Powder}       => {Onion}        0.6666667 0.8000000  1.200
{Onion}              => {Mirch Powder} 0.6666667 1.0000000  1.200
```

- In R

```r
# load required library
library(arules)

# Load the data from file
trans = read.transactions(file="code/smallData.csv", sep=",")

# inspect the data
inspect(trans)
```

```
##     items
## [1] {Bread,Milk,Mirch Powder,Onion}
## [2] {Bread,Mirch Powder,Onion}
## [3] {Bread,Milk}
## [4] {Milk,Mirch Powder,Onion}
## [5] {Mirch Powder}
## [6] {Bread,Milk,Mirch Powder,Onion}
```

```r
# observe support for all the items
itemFrequency(trans)
```

```
##        Bread         Milk Mirch Powder        Onion
##    0.6666667    0.6666667    0.8333333    0.6666667
```

```r
# Mine Rules
rulesT <- apriori(trans, parameter=list(support=0.5, confidence=0.7))
```

```
## Apriori
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

```
##
## Parameter specification:
##  confidence minval smax arem  aval originalSupport maxtime support minlen
##         0.7    0.1    1 none FALSE            TRUE       5     0.5      1
##  maxlen target   ext
##      10  rules FALSE
##
## Algorithmic control:
##  filter tree heap memopt load sort verbose
##     0.1 TRUE TRUE  FALSE TRUE    2    TRUE
##
## Absolute minimum support count: 3
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[4 item(s), 6 transaction(s)] done [0.00s].
## sorting and recoding items ... [4 item(s)] done [0.00s].
## creating transaction tree ... done [0.00s].
## checking subsets of size 1 2 3 done [0.00s].
## writing ... [17 rule(s)] done [0.00s].
## creating S4 object  ... done [0.00s].
```

```r
# Look at the summary
#summary(rulesT)
# See the rules
inspect(rulesT)
```

```
##      lhs                     rhs            support   confidence lift
## [1]  {}                   => {Mirch Powder} 0.8333333 0.8333333  1.000
## [2]  {Bread}              => {Milk}         0.5000000 0.7500000  1.125
## [3]  {Milk}               => {Bread}        0.5000000 0.7500000  1.125
## [4]  {Bread}              => {Onion}        0.5000000 0.7500000  1.125
## [5]  {Onion}              => {Bread}        0.5000000 0.7500000  1.125
## [6]  {Bread}              => {Mirch Powder} 0.5000000 0.7500000  0.900
## [7]  {Milk}               => {Onion}        0.5000000 0.7500000  1.125
## [8]  {Onion}              => {Milk}         0.5000000 0.7500000  1.125
## [9]  {Milk}               => {Mirch Powder} 0.5000000 0.7500000  0.900
## [10] {Onion}              => {Mirch Powder} 0.6666667 1.0000000  1.200
## [11] {Mirch Powder}       => {Onion}        0.6666667 0.8000000  1.200
## [12] {Bread,Onion}        => {Mirch Powder} 0.5000000 1.0000000  1.200
## [13] {Bread,Mirch Powder} => {Onion}        0.5000000 1.0000000  1.500
## [14] {Mirch Powder,Onion} => {Bread}        0.5000000 0.7500000  1.125
## [15] {Milk,Onion}         => {Mirch Powder} 0.5000000 1.0000000  1.200
## [16] {Milk,Mirch Powder}  => {Onion}        0.5000000 1.0000000  1.500
## [17] {Mirch Powder,Onion} => {Milk}         0.5000000 0.7500000  1.125
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

```r
# We can sort and see which are more important rules
 inspect(sort(rulesT, by="lift"))
```

```
##      lhs                      rhs            support   confidence lift
## [1]  {Bread,Mirch Powder} => {Onion}        0.5000000 1.0000000  1.500
## [2]  {Milk,Mirch Powder}  => {Onion}        0.5000000 1.0000000  1.500
## [3]  {Mirch Powder}       => {Onion}        0.6666667 0.8000000  1.200
## [4]  {Onion}              => {Mirch Powder} 0.6666667 1.0000000  1.200
## [5]  {Bread,Onion}        => {Mirch Powder} 0.5000000 1.0000000  1.200
## [6]  {Milk,Onion}         => {Mirch Powder} 0.5000000 1.0000000  1.200
## [7]  {Bread}              => {Milk}         0.5000000 0.7500000  1.125
## [8]  {Milk}               => {Bread}        0.5000000 0.7500000  1.125
## [9]  {Bread}              => {Onion}        0.5000000 0.7500000  1.125
## [10] {Onion}              => {Bread}        0.5000000 0.7500000  1.125
## [11] {Milk}               => {Onion}        0.5000000 0.7500000  1.125
## [12] {Onion}              => {Milk}         0.5000000 0.7500000  1.125
## [13] {Mirch Powder,Onion} => {Bread}        0.5000000 0.7500000  1.125
## [14] {Mirch Powder,Onion} => {Milk}         0.5000000 0.7500000  1.125
## [15] {}                   => {Mirch Powder} 0.8333333 0.8333333  1.000
## [16] {Bread}              => {Mirch Powder} 0.5000000 0.7500000  0.900
## [17] {Milk}               => {Mirch Powder} 0.5000000 0.7500000  0.900
```

- If you are interested in coming up with rules only for certain things

```r
rulesT <- apriori(trans, parameter=list(support=0.5, confidence=0.7),
                  appearance = list(rhs=c("Bread"), default="lhs"))
inspect(rulesT)
```

- Apriory property: Any subset of frequent itemset are also frequent : ABC if frequent - AB, AC, A, B, etc. are frequent
- Reverse is also true for infrequent - if A is infrequent AB can't be frequent
- Apriori stepwise

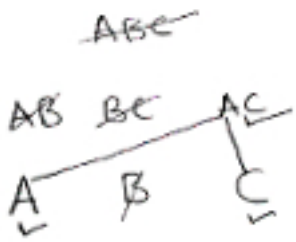
```
FrequentItemSet = {}
ItemSet = All items and their combinations in transaction table

LOOP (Till there are items in itemsets)
    1. Calculate support for every item in ItemSet
    2. Add those with support more than minimum support to FrequentItemSet
    3. ItemSet = Generate next level itemset using only the frequent items from step 1
LOOP_ENDS

GENERATE_RULES_USING FrequentItemSet > minConfidence
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

- Itemsets possible 2^(No. of items) - 1 = 2^3 = 8-1 = 7 : which means that many times we calculate support
- But the apriori property reduced that to = 4 only (i.e. support calculations for A, B, C and AC)
- Finding itemsets for a database with A,B,C items where A and C are frequent (above our minimum support) and B is not. Lattice of possible itemset :



- Limitation - Apriori needs to scan the transaction database for every level of lattice

1.3.3 FP-Growth Algorithm

- An efficient method to compute frequent itemset (Frequent Pattern)
- Needs scan of the database only 'twice'
- Follows 2 step approach
  - Create Frequent Pattern Tree of the itemsets
  - Generate frequent patterns from the tree
- FP-Tree Algorithm

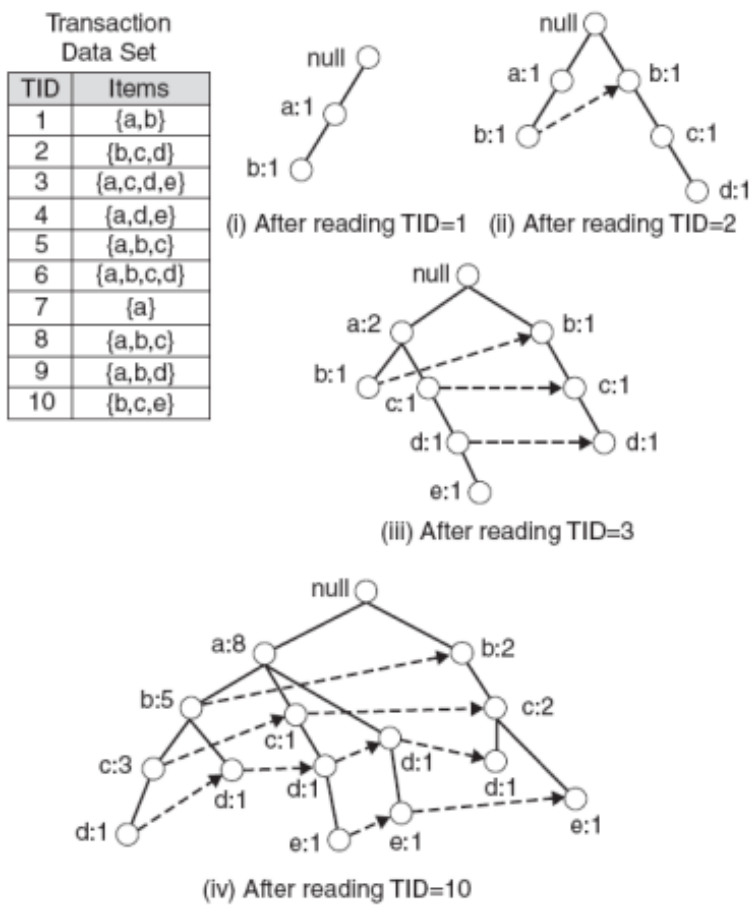
```
1. The Transactions:
b,a
b,d,c
a,c,d,e
a,d,e
a,b,c
a,b,c,d
a
a,b,c
a,b,d
b,c,e

2. Count Support - Pass 1
  a   b   c   d   e
  8   7   6   5   3

3. Arrange items according to their support - decreasing order - consider
only items above support, say 2
5. Create FP Tree for each item set - Pass 2
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

```
6. Generate Rules from FP-Tree
```



- Then conditional FP-Tree for each item is built using all possible paths to it
- For C it will be {a,c:1}, {b,c:2}, {a,b,c:3}
  - Rules - {a} => {c}, {b} => {c},{c} => {a},{a,b} => {c},{b,c} => {a}...
- For our minimum support of 2 : all above rules will hold true

1.4 Content Based Filtering

- Recommendations are based on content of the item and user's liking for individual content
- Uses data provided by user, either explicitly (rating) or implicitly (clicking on a link)
- The items are represented as per the content : As Vectors

```
        Romance   Action
DDLJ        1         0
Ravan       0         1
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

- For documents, or even for the movies, these vectors can be created using TF-IDF
  - TF-IDF is measure of importance, of a word to a document
  - Term Frequency (TF) is number of times the term occures in a document or movie labels/description
  - Inverse Document Frequency (IDF) is a measure of how common a term is across the documents - the, a, an, songs, fiction
  - IDF for 'a term' is calculated as - Total no. of documents/No. of documents with 'the term'
  - TF-IDF is a product of the two measures = TF*IDF for a term in a document - how native a terms is to the document
- User profile is also generated - Again a vector

```
        Romance   Action
Ram         1         1
Sita        1         0
Laxman      0         1
```

- User profile is generated using the movie ratings user has given and the movie/item features - a product of the two
- Based on these vectors the content is recommended using distance between item vector and user profile vector
- A 'simplified' content based recommendation:

```r
# Content Based filtering on a small dataset

# Movie Matrix with Features
M = c(0.2,0.6,0.2,0.8,0.2,0)
M<- matrix(M, nrow=2, byrow = T)
colnames(M) <- c("F1","F2","F3")
rownames(M) <- c("M1","M2")
M
```

```
##     F1  F2  F3
## M1 0.2 0.6 0.2
## M2 0.8 0.2 0.0
```

```r
# User rating matrix with movie ratings
U<- matrix(c(1,-1,-1,1), ncol = 2,byrow = T)
colnames(U) <- c("U1","U2")
rownames(U) <- c("M1","M2")
U
```

```
##    U1 U2
## M1  1 -1
## M2 -1  1
```


(c) 2017 CDAC, Mumbai. All Rights Reserved

```r
# Create User profile using rating matrix and the movie matrix -
# Dot product of the two matrix
UProfile <- t(M)%*%U
UProfile
```

```
##      U1   U2
## F1 -0.6  0.6
## F2  0.4 -0.4
## F3  0.2 -0.2
```

```r
# See what will be his liking for a new movie with following features
# A new movie vector with values for F1, F2 and F3
# Notice that the movie is more similar to the  M1
Mnew <- c(0.3,0.7,0)
# Cosine of the two vectors
library(lsa)
# for the first user
cosine(Mnew, UProfile[,1])
```

```
##           [,1]
## [1,] 0.1754656
```

```r
# for the second user
cosine(Mnew, UProfile[,2])
```

```
##            [,1]
## [1,] -0.1754656
```

- The distance vectors are calculated for a user for all movies and the top X movies are recommended
- Upsides - Purely content based and only individual user response is needed and easy to understand user profile
- Doesn't involve much computation and so is faster
- But can't recommend items outside these features

1.5 References

- Package arules documentation
- Data Mining and Predictive Analytics by by Daniel T. Larose and Chantal D. Larose

(c) 2017 CDAC, Mumbai. All Rights Reserved